

Ecopark Bicycle Parking

Requirement Analysis: Use-Case Analysis

Ecopark Bicycle Rental System Project Team

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Project Context and Rules

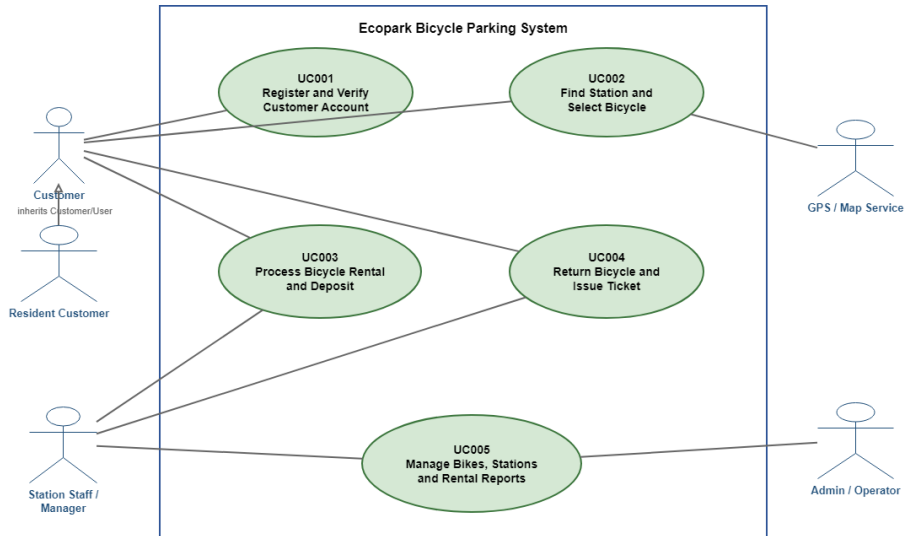
- Website for bicycle rental and return services for Ecopark visitors and residents.
- Manage stations, bicycle types, quantities, globally unique bicycle IDs, and bicycle status.
- First-time renters must create an account with email, ID document, address, and phone number.
- Bicycle pickup is handled at the station with Station Staff / Manager: VND 200,000 deposit + ID document.
- Rental prices: VND 50k/1 hour, VND 70k/2 hours, VND 100k/3 hours; late return: VND 30k/30 minutes.
- Residents with a valid Ecopark address and resident card receive a 40% rental discount.

Actor Model

Actor	Role
Customer/User	Regular renter: create account, find station, choose bicycle, pick up/return bicycle, view ticket/history.
Resident Customer	Specialized actor that inherits Customer/User; can submit resident card evidence for a 40% discount.
Station Staff / Manager	Station-level operations: check ID/deposit, hand over and receive bicycles, update bicycles at the station; Manager supervises station scope.
Admin / Operator	Central/system-wide operations: manage stations/bicycles/policies, review evidence, view statistics; Admin has higher configuration rights than Operator.
GPS / Map Service	External service that supports finding the nearest bicycle station.

Not actors: front-end, back-end, database, gate security.

Use-Case Overview



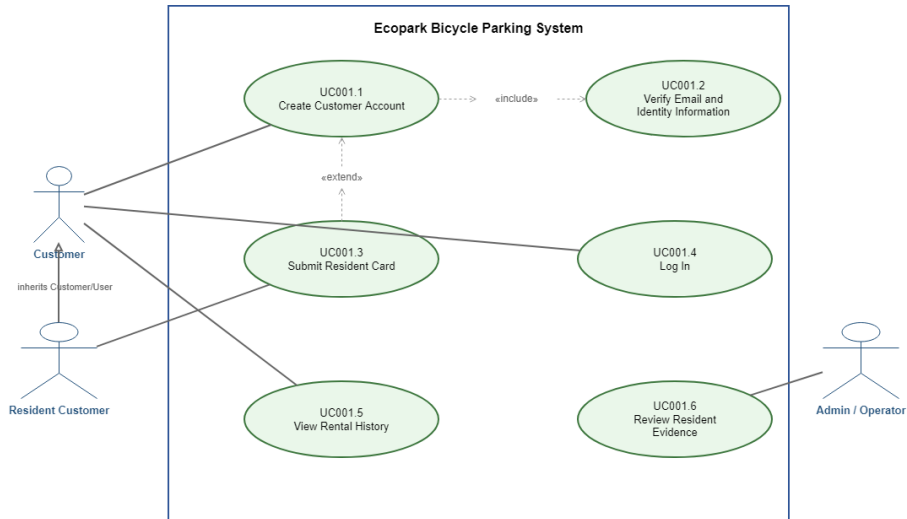
Relational Schema Summary

Area	Tables and relationships
Accounts	USERS → CUSTOMER_PROFILES; resident is customer_type=resident + RESIDENT_CARDS.
Fleet	STATIONS, BIKE_TYPES, BIKES, BIKE_STATUS_LOGS.
Request	CUSTOMER_PROFILES 1-n RENTAL_REQUESTS.
Rental lifecycle	RENTAL_REQUESTS 0..1-1 RENTALS; BIKES 1-n RENTALS.
Deposit and ticket	RENTALS 1-1 RENTAL_DEPOSITS; RENTALS 1-1 RENTAL_TICKETS.
Pricing	PRICING_POLICIES stores 50k/1h, 70k/2h, 100k/3h, 30k/30m and 40% discount.

UC001 - 1/2/3/5. Register and Verify Account

1. Code	UC001
2. Description	Customer creates an account, logs in, and provides identity information to become eligible for rental. Resident Customer may submit a resident card to request a 40% discount.
3. Actors	Customer; Resident Customer; Admin / Operator for manual review.
5. Preconditions	Valid email and ID document; account policy is configured; for discount requests, the customer has an Ecopark address and resident card.

UC001 - 4. Sub-Use Cases



- ❶ Customer chooses account creation or login.
- ❷ System displays the required fields.
- ❸ Customer enters credentials and personal information.
- ❹ System validates formats, duplicate data, and password.
- ❺ System creates the account/profile.
- ❻ Customer adds ID document information.
- ❼ System marks the account as rental eligible.
- ❽ If resident, customer submits address and resident card.
- ❾ System verifies or moves the evidence to manual review.
- ❿ System shows account, discount, and history status.

UC001 - 7. Alternative Flows

- Missing email/ID/phone/address: request completion.
- Email or phone already exists: block new account creation.
- Weak password: request re-entry.
- Invalid or blocked ID document: account is not rental eligible.
- Valid profile without resident proof: rent as regular customer.
- No resident card or customer skips it: no discount.
- Resident card mismatches or cannot be read: rejected or pending.
- Suspicious profile: send to Admin / Operator review.
- Locked account: block rental request creation.
- Rental history temporarily fails: login still allowed.

8. Input

- FullName, Email, PhoneNumber
- Password
- IdentityNumber, Address
- ResidentCardNumber if requesting 40%

9. Output

- CustomerID, AccountStatus
- RentalEligibility
- CustomerType
- DiscountStatus
- RentalHistorySummary

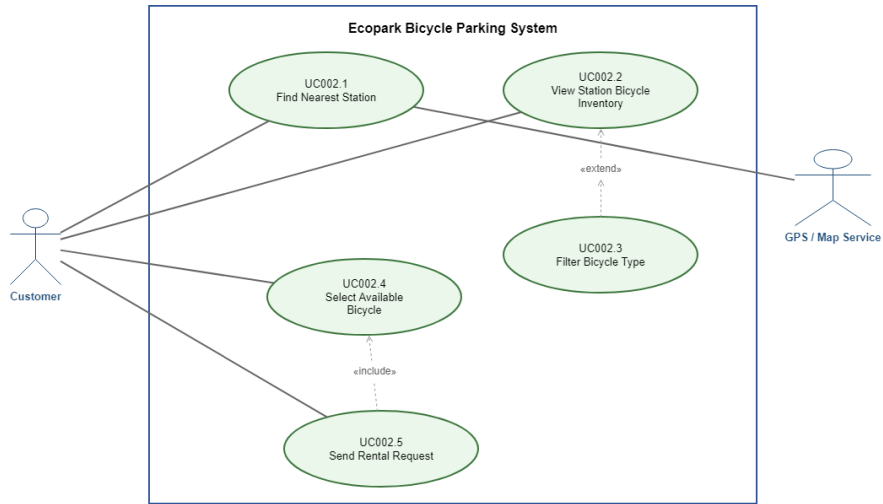
10. Rules

- Rental is blocked without valid email/identity.
- Active email must be unique.
- 40% discount applies only to verified residents.
- Rental history is tied to the account.

UC002 - 1/2/3/5. Find Station and Select Bicycle

1. Code	UC002
2. Description	Customer finds the nearest station, views available bicycles by type/preference, and sends a rental request before arriving at the pickup station.
3. Actors	Customer; GPS / Map Service.
5. Preconditions	Customer is logged in; account has identity information from UC001; station, bicycle, bicycle type, and current status data exist.

UC002 - 4. Sub-Use Cases



UC002 - 6. Basic Flow

- 1 Customer opens station search/rental feature.
- 2 System asks for location or search area.
- 3 GPS / Map Service returns location/area.
- 4 System lists nearest stations and available counts.
- 5 Customer selects a station to view details.
- 6 System shows bicycles, type, bicycle ID, and status.
- 7 Customer filters/selects an available bicycle.
- 8 System re-checks bicycle, station, and account.
- 9 Customer confirms the rental request.
- 10 System creates RentalRequest and pickup instructions.

UC002 - 7. Alternative Flows

- Location permission denied: enter location manually.
- GPS fails: use manual search.
- No nearby station: expand search radius.
- Station is suspended: suggest another station.
- Desired bicycle type is unavailable: change type or station.
- No ready bicycle: do not create request.
- Bicycle is just rented/broken: refresh and select again.
- Account lacks identity data: return to UC001.
- Customer cancels before sending: no request is created.
- Request expires: auto-cancel and return bicycle to available.

8. Input

- CustomerID
- Location/SearchArea
- StationID
- BicycleTypeFilter
- BicycleID

9. Output

- StationList
- BicycleAvailability
- RentalRequestID
- RequestStatus
- PickupInstruction

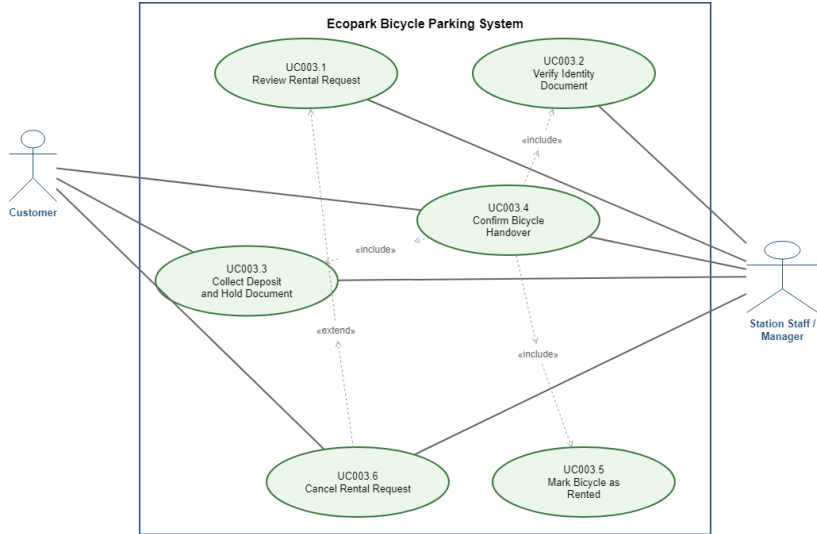
10. Rules

- Only available bicycles can be selected.
- Request is tied to Customer, Station, and Bicycle.
- Status must be re-checked before request creation.
- Account must be eligible for rental.

UC003 - 1/2/3/5. Process Rental and Deposit

1. Code	UC003
2. Description	Staff handles pickup at the station: review request, verify ID document, collect VND 200,000 deposit/hold document, and hand over the bicycle.
3. Actors	Customer; Station Staff / Station Manager.
5. Preconditions	RentalRequest is PendingPickup; Customer is at the correct station with ID document; bicycle is at the station and ready; Staff has permission.

UC003 - 4. Sub-Use Cases



- ❶ Customer presents the rental request code.
- ❷ Staff searches RentalRequest.
- ❸ System displays customer, bicycle, station, and duration.
- ❹ Staff checks ID card/document.
- ❺ Staff confirms the document matches the profile.
- ❻ Staff collects VND 200,000 deposit and records held document.
- ❼ System performs final bicycle/station status check.
- ❽ Staff hands over the correct bicycle to Customer.
- ❾ System creates Rental and deposit record.
- ❿ System marks bicycle as rented and shows receipt.

UC003 - 7. Alternative Flows

- Request not found/expired: create a new request.
- Request belongs to another station: do not process here.
- Customer forgot ID document: reject handover.
- Document mismatch: flag for review.
- Deposit below VND 200,000: do not create Rental.
- Customer refuses to leave document: reject handover.
- Bicycle is broken/needs repair: block handover.
- Bicycle is not at station: record data mismatch.
- Customer wants another bicycle: return to UC002.
- Rental/Deposit save fails: retry, do not mark rented.
- Receipt cannot print/display: Rental remains stored.

8. Input

- RequestID, StaffID
- IdentityDocumentType
- IdentityDocumentNumber
- DepositAmount
- DepositDocumentHeld

9. Output

- RentalID
- RentalStatus
- BicycleStatus
- StartedAt
- DepositReceipt
- PlannedReturnTime

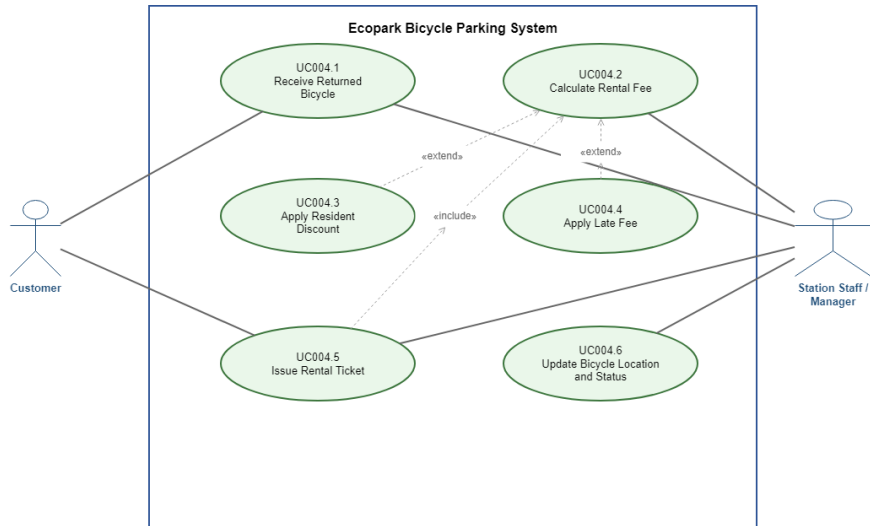
10. Rules

- No handover without valid deposit and document.
- Successful handover must create Rental.
- Each Request creates at most one successful Rental.
- StartedAt is used for UC004 fee calculation.

UC004 - 1/2/3/5. Return Bicycle and Issue Ticket

1. Code	UC004
2. Description	Customer returns the bicycle at any Ecopark station; Staff receives it; system updates location/status, calculates fees, applies discount/surcharge, and issues ticket.
3. Actors	Customer; Station Staff / Station Manager.
5. Preconditions	Customer has an InProgress Rental; bicycle is returned at an Ecopark station; Staff can receive bicycle; price and discount policies are active.

UC004 - 4. Sub-Use Cases



- ❶ Customer brings bicycle to an Ecopark station.
- ❷ Staff searches the Rental.
- ❸ System displays rental, bicycle, deposit, and time.
- ❹ Staff inspects bicycle condition.
- ❺ Staff confirms return station and return time.
- ❻ System updates return station and duration.
- ❼ System calculates base fee for 1/2/3-hour price.
- ❽ System applies 40% discount if resident is verified.
- ❾ System adds VND 30k/30 minutes if returned late.
- ❿ System issues ticket, stores history, and updates bicycle.

UC004 - 7. Alternative Flows

- Return station differs from pickup: accept and update location.
- Rental not found: send to manual review.
- Deposit mismatch: flag reconciliation.
- Bicycle broken/needs repair: mark broken, not rentable.
- Wrong station/time entered: allow correction before ticket.
- Actual duration below slot: follow pricing policy.
- Not resident/not verified: no 40% discount.
- Late time not exactly 30 minutes: round by policy.
- Late-fee dispute: store ticket and flag review.
- Ticket cannot print/send now: store for later view/print.
- Bicycle update fails: keep ticket and alert operations.

8. Input

- RentalID
- ReturnStationID
- ReturnedAt
- StaffID
- BicycleCondition
- ConditionNote

9. Output

- TicketID
- BaseFee, ResidentDiscount
- LateFee, TotalAmount
- BicycleLocation
- BicycleStatus

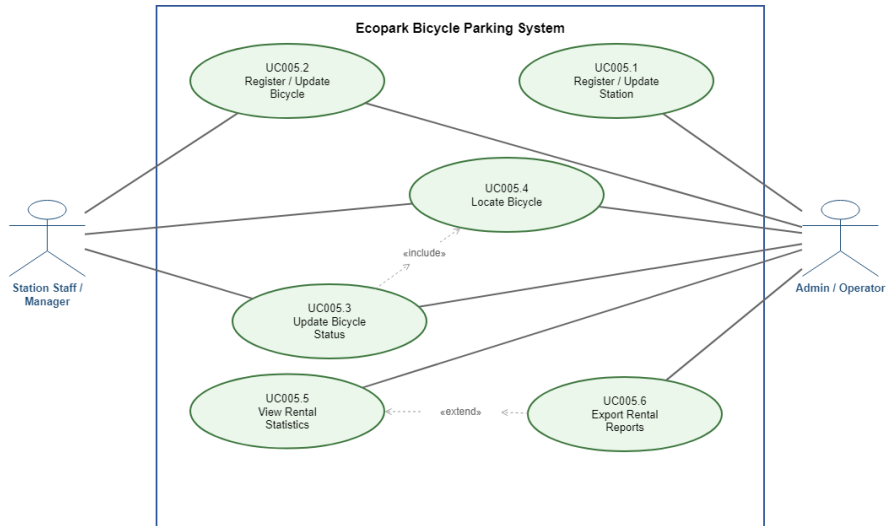
10. Rules

- Bicycle can be returned at any active Ecopark station.
- Ticket must include base fee, discount, surcharge, total.
- Completed Rental is added to Customer history.
- Broken bicycle must not appear in UC002.

UC005 - 1/2/3/5. Manage Bikes, Stations and Reports

1. Code	UC005
2. Description	Central operations: manage stations, bicycle types, bicycles, bicycle status/location, rental-return data, and statistics by day, week, month, and bicycle.
3. Actors	Admin / Operator; Station Staff / Station Manager.
5. Preconditions	Actor is logged in with proper permission; station/bicycle/rental data exists or is being created; system supports global and station-level permissions.

UC005 - 4. Sub-Use Cases



- ➊ Actor opens operations management.
- ➋ Actor chooses station, bicycle, status, locate, or statistics.
- ➌ Station: create/update name, address, coordinates, capacity, status.
- ➍ Bicycle: create/update bicycle ID, type, current station, status.
- ➎ Locate correct bicycle/location, then update actual status.
- ➏ System logs status/location change.
- ➐ Actor searches bicycle by ID or filters.
- ➑ Admin chooses report period and filters.
- ➒ System aggregates rentals, revenue, late fees, returns.
- ➓ System shows dashboard or exports report.

- Staff lacks global permission: limit to station scope.
- New station misses data: reject save.
- Capacity is below current bicycle count: reject reduction.
- Duplicate bicycle ID: block create/update.
- Bicycle is currently rented: block manual station change.
- Broken bicycle: remove from available list.
- Reopen broken bicycle: require inspection/repair note.
- Bicycle ID not found: do not change data.
- Empty report filters: show empty report.
- Data pending reconciliation: mark statistics provisional.
- Report export fails: keep filters and allow retry.

8. Input

- ActorID
- StationData
- BicycleData
- StatusChangeReason
- ReportPeriod
- ReportFilter

9. Output

- StationStatus
- BicycleStatus
- BicycleLocation
- StatusLogID
- RentalCount, RentalRevenue
- ReportFile

10. Rules

- Bike ID is unique across the system.
- Broken bicycle is not available for UC002.
- Staff is limited by permission scope.
- Reports support day, week, month, and bicycle.
- Every bicycle status change must be logged.

Presentation Checklist

- Start with overview diagram, actor hierarchy, and why 5 UCs were selected.
- For each UC, follow the same template: scope, sub-use cases, basic flow, alternative flows, data and rules.
- Use the imported draw.io diagrams as the visual reference for sub-use cases.
- Emphasize common mistakes avoided: no front-end/back-end/database actors; no action-only UCs.
- Use A4 PDF as SRS-style detailed backup for the complete tables.